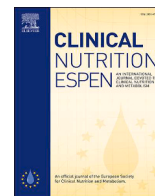




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Original article

Impact of transforming growth factor- β 2 enriched food for special medical purposes on nutritional status, microbiota and clinical outcomes after allogeneic hematopoietic stem cell transplantation: prospective analysis of 192 consecutive patients



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SUMMARY

Background and aims: Allogeneic hematopoietic stem cell transplantation (allo-HSCT) represents an essential treatment for various diseases despite being burdened by severe malnutrition, serious infections, graft-versus-host disease (GVHD) and gut microbiota dysbiosis. There is growing evidence suggesting that nutritional interventions may potentially lower these risks. In particular, nutritional supports enriched with the TGF- β 2 have shown promising benefits in Crohn's disease thanks to its immunomodulating and anti-inflammatory properties. Given the pathophysiological and clinical similarities between IBDs (Inflammatory Bowel Diseases) and gastrointestinal acute GVHD (GI aGVHD), this prospective study investigated the impact of a TGF- β 2 enriched Food for Special Medical Purposes (TGF- β 2 FSMP) on nutritional status, transplant outcomes, and microbiota composition in patients undergoing allo-HSCT.

Methods: Patients were divided into two groups according to their TGF- β 2 FSMP consumption defined by study protocol: Group A (Treatment Ratio ≥ 50 %, more than 50 % of prescribed FSMP assumed) and Group B (Treatment Ratio < 50 %). Transplant outcomes (Overall Survival, Transplant Related Mortality, Cumulative Incidence of acute GVHD, infectious complications, lymphocytes count 28 days after transplantation and microbiota composition) were analyzed according to TGF- β 2 FSMP consumption.

Results: A total of 192 consecutive patients who have undergone allo-HSCT were included, 112 in group A and 80 in group B. Adequate TGF- β 2 FSMP intake (Group A) significantly reduced severe malnutrition (PG-SGA C) at day +28 post-transplant, improved overall survival (OS) (median 42.2 months in Group A vs. 29.2 months in Group B, $p = 0.014$), and decreased non-relapse mortality (NRM) ($p = 0.018$). Furthermore, Group A exhibited a lower incidence of clinically significant acute graft-versus-host disease (aGVHD) (MAGIC II-IV) (18.4 % in Group A vs. 45.4 % in Group B, $p < 0.001$), particularly gastrointestinal aGVHD (12.8 % in Group A vs. 42.3 % in Group B, $p < 0.001$), and a reduced incidence of pneumonia (24.1 % in Group A, vs. 48.1 % in Group B, $p < 0.001$). Moreover, Group A showed significantly higher counts of NK lymphocytes (CD56+ and CD16+) at day +28 ($p = 0.012$ and $p = 0.002$, respectively). Microbiota analysis on 32 representative patients revealed no significant differences between groups after TGF- β 2 FSMP supplementation, while Firmicutes/Bacteroidetes (phyla commonly dominating gut microbiota) ratio showed a significant decrease in Group A at day +28 post-transplant compared to baseline ($p = 0.011$). Multivariable analysis identified adequate TGF- β 2 FSMP intake,

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absence of disease relapse, and lymphocyte count ≥ 140 cells/ μ L at day +28 as independent predictors of improved OS.

Conclusions: These findings highlight the potential of TGF- β 2 FSMP as a crucial nutritional intervention in allo-HSCT, positively influencing nutritional status, immune reconstitution, microbiota composition, and overall transplant outcomes. Further randomized trials are needed to confirm these data.

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1. Background

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a curative procedure for both several hematological disorders and non malignant diseases [1]. However, allo-HSCT is unfortunately associated with a large number of serious complications such as severe malnutrition, acute GVHD, infections and microbiota dysbiosis [2–5]. These complications can profoundly impact immune reconstitution, prolong hospitalization, reduce overall survival and lower patients' quality of life [6–8]. TGF- β 2 Food for Special Medical Purposes (TGF- β 2 FSMP) is a nutritional supplement commonly used in Gastroenterology for reducing the number and the severity of Inflammatory Bowel Diseases (IBD) flares, particularly in Crohn's disease [9–12]. The cytokine TGF- β 2, in fact, has various beneficial properties in these contexts, such as improving immunoregulation and promoting anti-inflammation [13]. The experimental use of TGF- β 2 FSMP in patients who have undergone allo-HSCT was based on the clinical and pathophysiological similarities (including the same pro-inflammatory cytokine networks) between IBD and gastro-intestinal (GI) aGVHD [14,15]. Furthermore, a recent study suggests that IBD and severe GVHD are both associated with certain genetic polymorphisms, such as those in the NOD2/CARD15 and TLR4 genes [16]. Different studies suggest that nutritional support has the potential to improve transplant outcomes by supporting immune functions, maintaining intestinal barrier integrity and microbiota composition and reducing infections and GVHD [17–19].

This study evaluated the potential benefits of TGF- β 2 FSMP on the prevention of severe malnutrition, non-relapse mortality (NRM) and overall survival (OS) and its role in modulating the gut microbiota, highlighting a complex interplay between nutrition, microbiota and immune function.

The primary objective of the study was to assess the impact of an adequate intake according to the study protocol (more than 50 % of the prescribed dose) of TGF- β 2 FSMP, administered as an experimental treatment, in preventing severe malnutrition (PG-SGA C) at +28 days post-transplant in a larger patient cohort [17,18].

Secondary objectives were to evaluate the role of nutritional support with TGF- β 2 FSMP and nutritional status on overall survival (OS), Non-Relapse Mortality (NRM) and on incidence of acute GVHD, particularly clinically significant acute GVHD (MAGIC stages II–IV), and gastrointestinal acute GVHD in a larger patient cohort and on microbiota composition.

2. Patients and methods

The study involved a total of 192 patients who underwent allogeneic hematopoietic stem cell transplantation (allo-HSCT) between April 2018 and August 2024.

2.1. Study design

Patients were divided into two groups based on their intake of TGF- β 2 FSMP nutritional support according to previous studies [17,18]:

- **Group A:** Patients with a Treatment Ratio (TR) ≥ 50 % (i.e., they consumed at least 50 % of the prescribed dose based on individual characteristics).
- **Group B:** Patients with a TR < 50 %.

Nutritional status was assessed by using the Patients Generated-Subjective Global Assessment (PG-SGA) questionnaire, widely used for malnutrition diagnosis in cancer patients [20–23]. Nutritional monitoring was conducted weekly until Day +28 post-transplant and performed by the institutional Dietitian.

2.2. Statistical analysis

Statistical analyses were conducted using SPSS Statistics (version 29.0.2.0), R (version 4.3.1) with RStudio (version 4.4.1), and GraphPad Prism (version 8.0.2). Clinical outcomes, including overall survival (OS), non-relapse mortality (NRM), and acute graft-versus-host disease (GVHD), were evaluated using the Log-Rank (Mantel-Cox) test, while OS was graphically represented through Kaplan-Meier survival curves. Comparisons between continuous variables were performed using the Mann-Whitney U test, whereas categorical variables were analyzed using Fisher's exact test. Microbiota diversity was assessed using both the Mann-Whitney U test and the Wilcoxon matched-pairs signed-rank test, as appropriate. A *p* of <0.050 was considered statistically significant.

2.3. Nutritional support and malnutrition

Nutritional support with a TGF- β 2-enriched food for special medical purposes (FSMP - Modulon-IBD, Nestlé) was administered from Day –7 to Day +28 post-transplant.

Modulon-IBD is a whole protein powdered formulation, 100 % casein based. It is composed of 42 % fats (25 % of the fat as medium chain triglycerides - MCT), 44 % carbohydrates, and 14 % proteins (extended composition reported as supplementary file). This polymeric food has a milk flavor and can be taken with cold or hot water and flavored with instant coffee, cocoa, etc., or it can be consumed mixed with ice cream or yogurt, or administered through enteral nutrition. The ingredients are: glucose syrup, milk protein, sucrose, milk fat, MCT oil, minerals (magnesium chloride, calcium phosphate, sodium citrate, potassium citrate, potassium hydroxide, calcium carbonate, potassium chloride, manganese sulphate, ferrous sulphate, zinc sulphate, copper sulphate, sodium molybdate, chromium chloride, potassium iodide, sodium selenate), corn oil, emulsifier (soy lecithin), vitamins (C, E, niacin, pantothenic acid, B6, thiamin, A, riboflavin, folic acid, K, biotin, D, B12), choline bitartrate.

Dosages were individualized based on each patient's Body Mass Index (BMI) and Total Daily Energy Expenditure (TDEE), the latter calculated from Basal Metabolic Rate (BMR). Caloric needs from TGF- β 2 FSMP were adjusted according to BMI categories and nutritional status, as defined by PG-SGA classification (A, B, or C), with higher caloric targets assigned to patients with poorer nutritional status or lower BMI (Table 1). The proportion of the

Table 1Calories recommended by TGF- β 2 FSMP according to TDEE, BMI and PG-SGA.

	PG-SGA A	PG-SGA B	PG-SGA C	Calories by TGF- β 2 FSMP of TDEE
If BMI <20	40 kcal/kg	42 Kcal/Kg	44 kcal/kg	20 %
If BMI 20–24.9	35 kcal/kg	39 kcal/kg	42 kcal/kg	15 %
If BMI 25–29.9	30 kcal/kg	33 kcal/kg	36 kcal/kg	12 %
If BMI >30	25 kcal/kg	28 kcal/kg	30 kcal/kg	10 %
		TDEE		

TDEE covered by the FSMP ranged from 10 % to 20 %, depending on BMI. Based on the actual intake relative to the recommended dose, patients were stratified into two groups: Group A ($n = 112$), who received ≥ 50 % of the prescribed caloric target, and Group B ($n = 80$), who received < 50 %.

All participants in the study followed a standard hospital diet, which provided approximately 1800 kcal per day. This diet included three main meals: breakfast, lunch, and dinner. Specifically, breakfast consisted of tea, milk, or barley coffee served with biscuits, brioches, or rusks. Lunch and dinner were composed of a first course, a second course, and a side dish, with 4–5 choices available for each course. Additionally, an optional afternoon snack was available to all patients and included ice cream, yogurt, pudding, crackers, breadsticks, fruit juices, or soft drinks.

The enrolled patients were categorized based on the type of parenteral nutrition received, either total (TPN) or partial (PPN). The total parenteral nutrition dose was calculated according to the patient's nutritional requirements, while the partial parenteral nutrition dose was determined based on the amount of nutrients orally consumed by the patient.

2.4. Nutritional status, outcomes, and immune analysis

To investigate the relationship between clinical and laboratory parameters and post-transplant outcomes, patients were assessed for nutritional status, microbiota composition, and immune profile. Nutritional status was monitored weekly up to Day +28 post-stem cell infusion using the Patient-Generated Subjective Global Assessment (PG-SGA), with malnutrition defined as a PG-SGA score of category C or ≥ 9 at Day +28. Immune reconstitution was evaluated by flow cytometry (Becton-Dickinson FACSCanto II), measuring lymphocyte subpopulations at Day +28. Two antibody panels were employed: a TBNK panel for basic immunophenotyping (CD3, CD4, CD8, CD45, CD19, CD16, and CD56) and an alloHSCt panel for T cell subpopulations (CD45RA, CD45RO, CD25, and CD127). Acute graft-versus-host disease (GVHD) was diagnosed based on both clinical presentation and histopathological confirmation according to MAGIC Criteria [24].

2.5. Microbiota collection and analysis

Pre-transplant (baseline, Day –7) and post-transplant (Day +28) gut microbiota analysis was performed using stool samples. Stool samples were collected in the morning using rectal swabs (FecalSwab™ system, Copan Italia, Brescia, Italy) that had been kept in the refrigerator for less than 12 h and subsequently frozen at -80 °C.

Stool samples were stored at -80 °C in the Microbiology and Virology Laboratory of Spedali Civili di Brescia until microbial DNA extraction, which was performed according to the manufacturer's instructions using the QIAamp DNA Mini Kit (Qiagen Retsch GmbH, Hannover, Germany). A negative control, in which stools were replaced with molecular biology grade water, was extracted at the same time with the patient samples.

Before library preparation, bacterial DNA concentration and integrity in samples were analyzed by qualitative real-time PCR in SYBR Green chemistry targeting the 16 S rRNA gene, and visually by electrophoresis on 1 % agarose gel containing SYBR Safe. The concentration and purity of the extracted DNA were also assessed using Qubit 4 Fluorometer (Thermo Fisher Scientific).

Microbiota composition was assessed by V3–V4 16 S rRNA regions amplification, following 16 S Metagenomic Sequencing Library Preparation protocol (Illumina) according to the manufacturer's instructions. For paired-end sequencing, MiSeq Reagent Kit v2 (500-cycles) on Illumina MiSeq platform was used.

Following sequencing, raw reads were processed using Trimmomatic (version 0.39) to remove primers and adapters. Low-quality reads were filtered out, and the remaining sequences were clustered into operational taxonomic units (OTUs), which were subsequently assigned to taxonomic ranks using reference databases within the R environment. Samples with fewer than 50,000 high-quality reads were excluded from further analysis. Compositional and functional profiling of the microbiota, along with comparative and statistical analyses, were performed using the *microbiome* package in R (version 4.3.1 with RStudio version 4.4.1) and GraphPad Prism (version 8.0.2). This comprehensive analytical approach aimed to explore the potential impact of nutritional support and immune reconstitution on post-transplant recovery, with a particular focus on microbiota diversity. The Firmicutes/Bacteroidetes (F/B) ratio was calculated to describe the balance between two major groups of bacteria in the gut microbiota. Variations in this ratio are linked to different health conditions, particularly in IBD [25].

2.6. Ethical considerations

The study was reviewed and approved by the Ethics Committee of Spedali Civili di Brescia, and all patients provided informed consent according to the Declaration of Helsinki.

3. Results

3.1. Demographic characteristics

One hundred ninety-two patients were prospectively analyzed and male/female ratio was 62/38 %. Median age was 58 years for males (range 18–78) and females (range 22–72) and median follow-up was 12 months (range 1–45 months).

One hundred and forty-three patients (74.5 %) were diagnosed with acute myeloid leukemia, myelodysplastic syndrome, or myeloproliferative disorders; 46 patients (24 %) with lymphoproliferative diseases and 3 patients (1.5 %) with severe immunodeficiency. At the time of conditioning, 72 patients (37.5 %) were in complete remission, 89 (46.4 %) in partial remission, and 31 (16.1 %) had active or advanced disease. The donor was a match-related (HLA-identical sibling) in 56 cases (29.2 %), a matched unrelated donor (MUD) in 90 cases (46.8 %), and a haploidentical donor in 46 cases (24 %). The main graft source was peripheral blood (94.3 %) followed by bone marrow (6.7 %). Conditioning

regimens included myeloablative conditioning (MAC) in 95 patients (49.5 %) and reduced-intensity conditioning (RIC) in 97 patients (50.5 %). According to the Hematopoietic Cell Transplantation-Comorbidity Index (HCT-CI), patients had a median score of 3, consistent with a high-risk comorbidity profile.

3.2. Impact of TGF- β 2 FSMP on nutritional status

At day -7 from transplant, the first nutritional assessment using the PG-SGA tool classified patients as follows: 146 (76 %) in good nutritional status (PG-SGA A), 43 (22.4 %) in fair nutritional status (PG-SGA B), and 3 (1.6 %) in malnourished status (PG-SGA C). At day $+28$ post-transplant, a re-evaluation found that 114 patients (59 %) were non-malnourished (PG-SGA A or B), while 78 (41 %) had severe malnutrition (PG-SGA C). The intake of the TGF- β 2 FSMP nutritional supplement was monitored during hospitalization. The Treatment Ratio (TR), reflecting the dose taken relative to the prescribed dose, had a mean of 66 % and a median of 71 %, with a range of 0 %–224 %. Patients were divided into two groups: Group A (TR ≥ 50 %) with 112 patients, and Group B (TR < 50 %) with 80 patients, according to the first published study rules [17,18]. Mean group A TR was 93 ± 2.5 % in comparison to group B with a mean of 22 ± 1.7 % ($p < 0.001$). Poor intake was mainly due to disgust and unpleasant taste, while other causes included mucositis, GVHD, and invasive ventilation. No statistically significant differences were observed between the two groups demographic, clinical, hematological, and transplant-related characteristics (Table 2). The impact of TGF- β 2 FSMP nutritional support was evaluated based on the nutritional status of patients at $+28$ days post-HSC transplant as previously reported [9]. Notably, 87 patients (77.6 %) in Group A were well nourished or moderately malnourished (PG-SGA A-B) at $+28$ days, compared to only 27 patients (33.8 %) in Group B. Conversely, the incidence of poor nutritional status (PG-SGA C) was only 22.4 % in patients with adequate TGF- β 2 FSMP intake, compared to 63.6 % in Group B. This

result is statistically significant ($p < 0.001$), meets the primary endpoint of the original study and confirms the previously reported results [17,18]. For the high nutritional risk definition based on a PG-SGA score ≥ 9 , similar results were observed: 58 % of Group A patients had this score at $+28$ days versus 90 % in Group B ($p < 0.001$). A direct linear correlation between TR and numeric PG-SGA was confirmed ($R^2 = 0.136$, $p < 0.001$): the higher the FSMP intake (in terms of TR percentage), the lower the numerical values of PG-SGA (Fig. 1).

Among all patients included in the study, 36 (32.2 %) patients in Group A received parenteral nutrition, with 21 (18.8 %) receiving PPN and 15 (13.4 %) receiving TPN. In Group B, 46 (57.5 %) patients received parenteral nutrition, with 12 (15 %) on PPN and 34 (42.5 %) on TPN. The difference observed was statistically significant between the two groups (A and B) ($p < 0.001$).

3.3. Impact of TGF- β 2 FSMP on acute GVHD

With regard to the association between TGF- β 2 FSMP and acute GVHD, in particular clinically significant acute GVHD (MAGIC Stage II–IV) and gastrointestinal acute GVHD, the following findings emerged from the analysis: the cumulative incidence of acute GVHD of any grade was higher in patients with lower intake (Group B), at 53.6 %, compared to 38.3 % in those with adequate intake (Group A), although the difference did not reach statistical significance ($p = 0.065$; data not shown). In contrast, the incidence of clinically significant acute GVHD (MAGIC stages II–IV) was markedly higher in Group B (45.4 %) than in Group A (18.4 %), with a highly significant difference ($p < 0.001$; Fig. 2). Similarly, the cumulative incidence of gastrointestinal acute GVHD was substantially greater in Group B (42.3 %) compared to Group A (12.8 %) ($p < 0.001$; Fig. 3). Notably, no patients in Group A died from acute GVHD, while only one death attributable to acute GVHD occurred in Group B (not statistically significant).

Table 2

Population characteristics according to Group of TGF- β 2 FSMP assumption; AML/MDS: acute Myeloid Leukemia or Myelodysplastic Syndrome, LPD: Lymphoproliferative Disorders; MUD: Matched Unrelated Donor, MAC: Myeloablative Conditioning, RIC: Reduced Intensity Conditioning, PG-SGA: Patients Generated-Subject Global Assessment.

Population Characteristics	Total: n (%)	TGF- β 2 FSMP group A	TGF- β 2 FSMP group B	p
Sex: n (%)				
Male	119 (62 %)	68 (60.7 %)	51 (63.8 %)	0.669
Female	73 (38 %)	44 (39.3 %)	29 (36.2 %)	
Age:				
Median	58.5	59	56	0.110
Range	18–78	20–78	18–75	
Diagnosis: n (%)				
AML/MDS	135 (70.3 %)	83 (74.1 %)	52 (65 %)	0.346
LPD	52 (27.1 %)	27 (24.1 %)	25 (31.3 %)	
Others	5 (2.6 %)	2 (1.8 %)	3 (3.7 %)	
State of disease at the admission: n (%)				
Complete Remission	72 (37.5 %)	42 (37.5 %)	30 (37.5 %)	0.900
Minimal Residual Disease	89 (46.4 %)	53 (47.3 %)	36 (45 %)	
Uncontrolled disease	31 (16.1 %)	17 (15.2 %)	14 (17.5 %)	
Donor type: n (%)				
Related	56 (29.2 %)	36 (32.1 %)	21 (26.3 %)	0.317
MUD	90 (46.8 %)	54 (48.2 %)	36 (45 %)	
Haploidentical	46 (24 %)	22 (19.7 %)	23 (28.7 %)	
Conditioning: n (%)				
MAC	95 (49.5 %)	56 (50 %)	39 (48.8 %)	0.864
RIC	97 (50.5 %)	56 (50 %)	41 (51.2 %)	
Source of HSC: n (%)				
Peripheral Blood	181 (94.3 %)	104 (92.9 %)	77 (96.3 %)	0.319
Bone Marrow	11 (5.7 %)	8 (7.1 %)	3 (3.7 %)	
PG-SGA class at admission: n (%)				
PG-SGA A	146 (76 %)	86 (76.8 %)	61 (76.3 %)	0.110
PG-SGA B	43 (22.4 %)	26 (23.2 %)	16 (20 %)	
PG-SGA C	3 (1.6 %)	0 (0 %)	3 (3.7 %)	

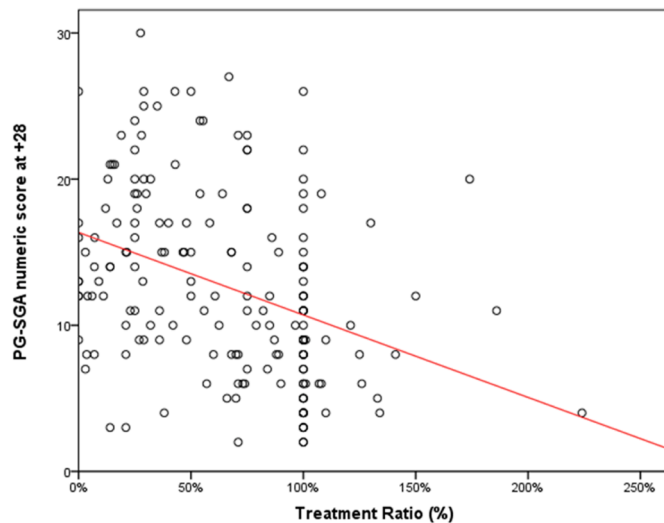


Fig. 1. Relationship between PG-SGA global score and Treatment Ratio (TR) of TGF- β 2 FSMP.

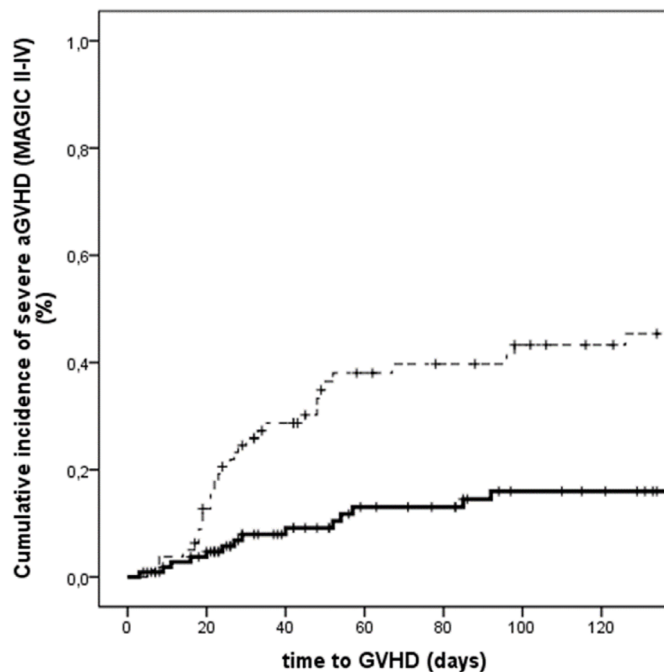


Fig. 2. CI of severe acute GVHD (MAGIC II-IV) according to TGF- β 2 FSMP treatment group: Group A continuous line, Group B dotted line.

Furthermore, it was found that patients who received TPN developed a significantly higher cumulative incidence of gastrointestinal GVHD (of any grade) compared to those who received PPN or no parenteral nutrition (42.5 % vs 13.4 %, $p = 0.003$).

3.4. TGF- β 2 FSMP, lymphocyte subpopulations and infections

The evaluation of the impact of TGF- β 2 FSMP on lymphocyte subpopulations (particularly on NK cells) is a key endpoint of this study. As shown in Table 3, analysis of lymphocyte subpopulations at Day +28 post-transplant revealed significantly higher counts in patients with adequate TGF- β 2 FSMP intake (Group A) compared to those with lower intake (Group B). Total lymphocyte counts

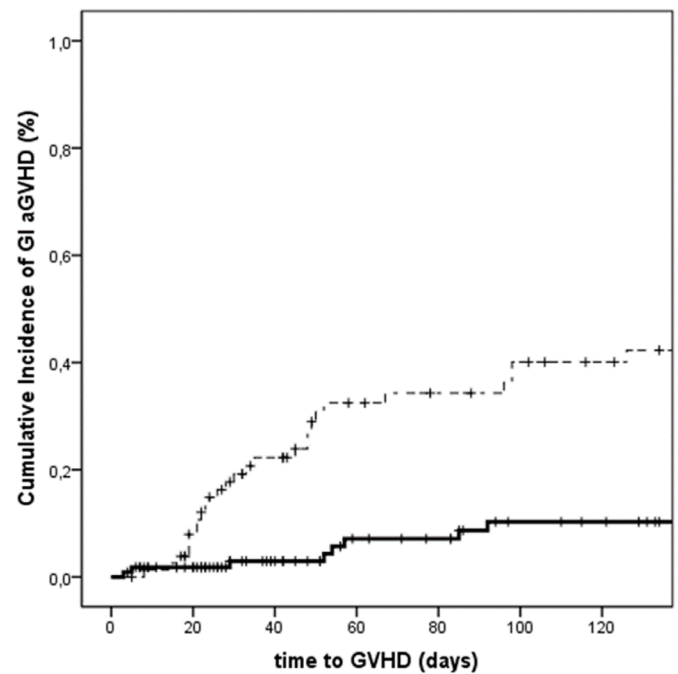


Fig. 3. CI of acute gastrointestinal GVHD according to TGF- β 2 FSMP treatment group: Group A continuous line, Group B dotted line.

were markedly increased in Group A, with a mean of 460.4 ± 44.4 cells/ μ L, compared to 333.8 ± 43.8 cells/ μ L in Group B ($p = 0.001$). Similarly, T-helper lymphocytes ($CD3^+CD4^+$) were more abundant in Group A (100.4 ± 15.1 cells/ μ L) than in Group B (63.1 ± 11.1 cells/ μ L; $p = 0.020$). Natural killer (NK) cell populations also differed significantly between groups: $CD56^+$ NK cells were higher in Group A (155.3 ± 16.3 cells/ μ L) versus Group B (116.7 ± 23.6 cells/ μ L; $p = 0.012$), as were $CD16^+$ NK cells (130.1 ± 15.6 vs. 76.9 ± 17.1 cells/ μ L; $p = 0.002$). The number of T-reg Lymphocytes was not statistically different between the two groups (7.58 ± 0.97 cells/ μ L vs 7.90 ± 0.86 cells/ μ L; $p = 0.417$).

The main short-term infectious complications in the post-transplant period can be summarized into four categories: sepsis, pneumonia, enteritis and fever of unknown origin (FUO). In this context, the following associations were observed:

- The incidence of sepsis was not related to the level of TGF- β 2 FSMP intake, with an incidence of 50.6 % in Group B and 45.5 % in Group A ($p = 0.487$).
- Pneumonia showed a higher frequency in Group B, with an infective lung involvement of 48.1 % in the group not treated with an adequate dose of TGF- β 2 FSMP (Group B), compared to 24.1 % of patients in Group A, with a statistically significant value ($p < 0.001$).
- The number of cases of enteritis, although higher in Group B with an incidence of 25.3 % compared to 18.8 % observed in Group A, was not statistically significant between the two TGF- β 2 FSMP intake groups ($p = 0.284$).

3.5. TGF- β 2 FSMP, GVHD and microbiota analysis

3.5.1. Microbiota composition and Firmicutes/Bacteroidetes ratio

Microbiota analyses were performed on 32 patients from the overall cohort, using fecal samples collected at day -7 (pre-HSCT) and day +28 (post-HSCT). Among them, 22 patients were in Group

Table 3Comparison of mean values of lymphocyte subpopulations based on TGF- β 2 FSMP Group.

Lymphocyte subpopulations	Group A (mean values)	Group B (mean values)	p-value
Lymphocytes at +28 days (cells/ μ L)	460.4 $\pm$ 44.4	333.8 $\pm$ 43.8	0.001
T-Helper Lymphocytes CD3+CD4+ at +28 (cells/ μ L)	100.4 $\pm$ 15.1	63.1 $\pm$ 11.1	0.020
NK Lymphocytes CD56+ at +28 (cells/ μ L)	155.3 $\pm$ 16.3	116.7 $\pm$ 23.6	0.012
NK Lymphocytes CD16+ at +28 (cells/ μ L)	130.1 $\pm$ 15.6	76.9 $\pm$ 17.1	0.002

A and 10 in Group B. The subset was representative of the entire population, showing no statistically significant differences in demographic or clinical features (Table 4). Gut microbiota analyses at baseline (pre-transplant period) reveal no significant differences in bacterial families between the two TGF- β 2 FSMP intake groups. Post-transplant analysis revealed in Group A a higher, but not significant, relative prevalence of *Clostridiaceae*, *Flavobacteriaceae*, *Streptococcaceae*, *Lachnospiraceae*, and *Ruminococcaceae* compared to Group B. Conversely, Group B shows a greater relative abundance of *Xanthomonadaceae*, *Pseudomonadaceae*, *Veillonellaceae*, *Prevotellaceae*, and *Bacteroidaceae* (Fig. 1 appendix). One patient of Group A and one patient of Group B were excluded from analysis because of a low number of quality reads obtained from sequencing (<50,000).

The F/B ratio in two TGF- β 2 FSMP intake groups was analyzed. Group A showed a lower post-HSCT F/B ratio and the difference was statistically significant ($p = 0.011$) while Group B showed a higher post-HSCT F/B ratio but the difference was not statistically significant (Fig. 4).

3.5.2. Microbiota and occurrence of GI-aGVHD

Results of pre-transplant microbiota analysis show a not significant difference in the relative abundance of major bacterial populations between patients who will develop GI-aGVHD of any severity and those who will not (data not shown). Despite not significant, patients that developed or will develop acute GI-GVHD

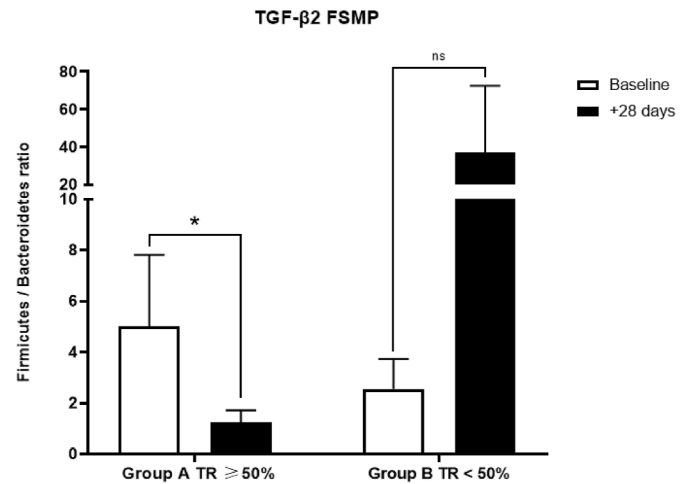


Fig. 4. Firmicutes/Bacteroidetes ratio according to TGF- β 2 FSMP Treatment Ratio at baseline and after transplantation. * p -value = 0.011, Wilcoxon matched-pairs signed rank test. Data are shown as mean $\pm$ SEM.

of any severity show, at day +28 post-transplant, a higher relative abundance of *Ruminococcaceae*, *Enterobacteriaceae*, *Xanthomonadaceae* and *Bacteroidaceae*. Five of the 6 patients who developed acute GI-GVHD belonged to Group B. Conversely, *Lachnospiraceae*, *Flavobacteriaceae*, *Clostridiaceae* and *Veillonellaceae* are more

Table 4

Population characteristics: comparison between the whole cohort and patients who underwent microbiome analysis pre- and post-transplant; AML/MDS: acute Myeloid Leukemia or Myelodysplastic Syndrome, LPD: Lymphoproliferative Disorders; MUD: Matched Unrelated Donor, MAC: Myeloablative Conditioning, RIC: Reduced Intensity Conditioning.

Population Characteristics	Total cohort: n (%)	Patients with analyzed microbiota	p value
Sex: n (%)			
Male	119 (62 %)	18 (56.3 %)	0.538
Female	73 (38 %)	14 (43.7 %)	
Age:			
Median	58.5	57	0.679
Range	18–78	18–74	
Diagnosis: n (%)			
AML/MDS	143 (74.5 %)	26 (81.3 %)	0.610
LPD	46 (24 %)	6 (18.7 %)	
Others	3 (1.5 %)	0 (0 %)	
State of disease at the admission: n (%)			
Complete Remission	72 (37.5 %)	12 (37.5 %)	0.997
Minimal Residual Disease	89 (46.4 %)	15 (46.9 %)	
Uncontrolled disease	31 (16.1 %)	5 (15.6 %)	
Donor type: n(%)			
Related	56 (29.2 %)	10 (31.3 %)	0.941
MUD	90 (46.8 %)	18 (56.3 %)	
Haploidentical	46 (24 %)	4 (12.4 %)	
Conditioning: n (%)			
MAC	95 (49.5 %)	16 (50 %)	0.956
RIC	97 (50.5 %)	16 (50 %)	
Source of HSC: n (%)			
Peripheral Blood	181 (94.3 %)	31 (96.9 %)	0.545
Bone Marrow	11 (5.7 %)	1 (3.1 %)	
TGF-β2 FSMP group: n (%)			
Group A	112 (58.9 %)	22 (68.8 %)	0.657
Group B	80 (41.1 %)	10 (31.2 %)	

abundant in those who will not experience this complication within the first 100 days post-transplant (Fig. 2 appendix).

3.6. Impact of TGF- β 2 FSMP on OS

Estimated mean Overall Survival (OS) of the entire cohort is 30.5 months (CI: 27.5–33.48 months). According to the TGF- β 2 FSMP assumption the difference was proven significant ($p = 0.014$, Group A median 42.2 months, Group B 29.2 months, Fig. 5).

3.7. Impact of TGF- β 2 FSMP on NRM and relapse

For the entire available follow-up period, with a median of 12 months (range: 1–45 months), non-relapse mortality was lower in patients with adequate intake of TGF- β 2 FSMP (Group A) [10 % (CI: 5.7–14.2 %) in group A and 25.2 % (CI: 19.1–31.4 %) in group B, $p = 0.018$, Fig. 6]. Mean Hospital Stay was significantly lower in Group A than in Group B (30.6 days vs 37.6 days, $p = 0.004$).

3.8. Multivariable analysis

In the multivariable analysis using Cox regression model, the variables impacting overall survival were as follows: TGF- β 2 FSMP intake Group A (TR ≥ 50 %) ($p = 0.039$, HR = 0.54 [0.31–0.97]), disease relapse ($p < 0.001$, HR = 4.09 [2.34–7.17]), and lymphocyte count ≥ 140 cells/ μ L on day +28 post-transplant ($p = 0.010$, HR = 0.46 [0.26–0.83]).

4. Discussion

This study confirmed that adequate intake of TGF- β 2 FSMP is significantly associated with a reduced incidence of malnutrition 28 days after allogeneic hematopoietic stem cell transplantation (allo-HSCT).

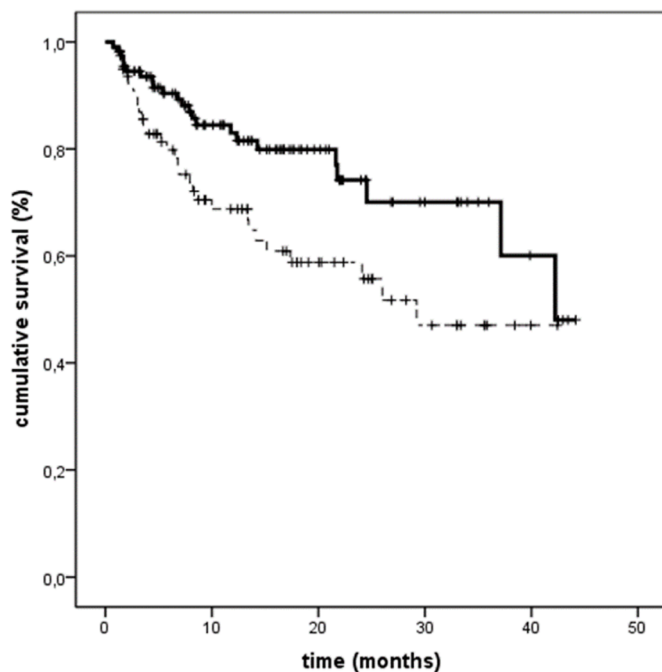


Fig. 5. Overall Survival (OS) according to TGF- β 2 FSMP treatment group, Group A continuous line, Group B dotted line.

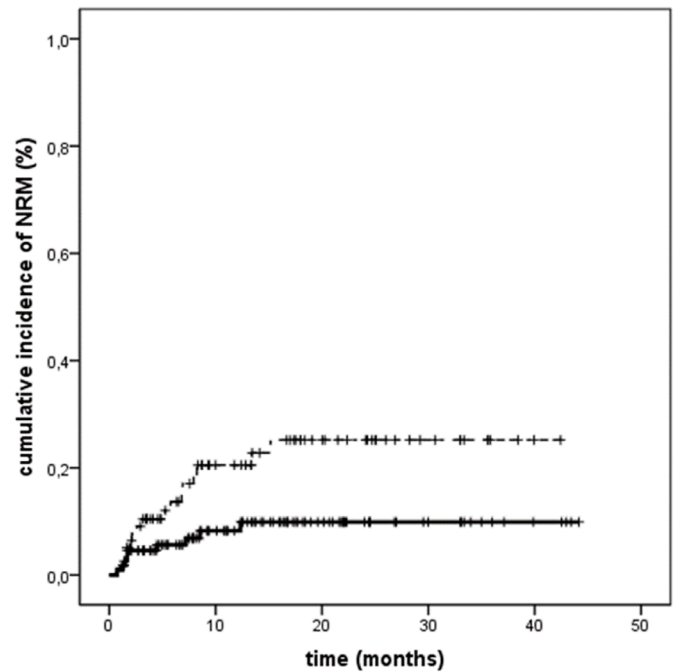


Fig. 6. NRM according to TGF- β 2 FSMP treatment group: Group A continuous line, Group B dotted line.

4.1. Impact on malnutrition

Malnutrition was defined through the Patient-Generated Subjective Global Assessment (PG-SGA) tool, specifically as Category C or a PG-SGA score ≥ 9 . The TGF- β 2 FSMP nutritional support seems to be associated with a potential protective effect, hypothetically preventing malnutrition by maintaining nutritional reserves: A potential effect on the prevention of malnutrition is due to the fact that all patients who received an adequate dose of Modulen-IBD started treatment from the beginning of the conditioning phase and while still in good nutritional status. In clinical practice, nutritional intervention usually begins when a patient has failed to consume an adequate number of calories for at least three days. However, in this case, by starting nutritional support early, the intervention can be considered pre-emptive aligned with the concept that all patients undergoing alloHSCT are at risk of severe malnutrition. This hypothesis should be tested in a controlled study. The data regarding the use of parenteral nutrition seem to confirm the impact of malnutrition on transplant outcomes, particularly on gastrointestinal GVHD.

4.2. Impact on transplant outcomes

The cytokine TGF- β 2 likely contributed to these outcomes by promoting epithelial cell trophism and immune modulation. This underscores the interplay between nutritional status and gut health, with implications for reducing complications such as enteritis and pneumonia. Notably, TGF- β 2 FSMP intake supported higher lymphocyte subpopulation counts, particularly NK cells (CD56+ and CD16+), demonstrating its role in both maintaining immune functionality and reducing transplant-related complications [26]. The study observed a superior overall survival (OS) rate in patients receiving adequate TGF- β 2 FSMP dosages (TR ≥ 50 %) compared to those with insufficient intake. Multivariate analysis confirmed this association, identifying adequate nutritional support as a protective factor against critical post-transplant

complications. Adequate TGF- β 2 FSMP intake was linked to favorable immune recovery, particularly in lymphocyte and NK cell counts, further correlating with improved OS. These findings align with previous research demonstrating the prognostic significance of early immune reconstitution, especially NK cells, in preventing infections, GVHD, and relapse [27–30]. Furthermore, the cytokine TGF- β 2 has been implicated in supporting T-reg cell proliferation (even if, in our study, we did not find any significant difference in the number of T-reg between group A and Group B at +28 days) and intestinal epithelial cell health [31–35], which may explain its survival benefits. While the study found no significant differences in sepsis incidence, the observed reduction in gastrointestinal and pulmonary complications contributed to enhanced OS, highlighting the critical role of nutrition in allo-HSCT outcomes. Non-relapse mortality (NRM) was significantly higher in patients with insufficient TGF- β 2 FSMP intake. Adequate nutritional support was inversely correlated with NRM, underscoring its protective role against severe complications such as GI-GVHD and infections. Relapse rates, although not significantly addressed in this study, were observed to correlate with lymphocyte counts and nutritional status. Maintaining intestinal barrier integrity through sufficient TGF- β 2 FSMP intake likely played a role in reducing bacterial translocation and subsequent systemic complications [14], which are major contributors to NRM. Additionally, patients with better nutritional status displayed lower instances of pneumonia and enteritis, further strengthening the evidence that TGF- β 2 FSMP helps mitigate infection-related mortality. This protective effect is supported by TGF- β 2's ability to foster immune tolerance and support intestinal health, providing a mechanistic basis for the observed improvements in NRM [31–34,36]. The intake of TGF- β 2 FSMP at sufficient dosages (TR $\geq$ 50 %) significantly reduced the incidence of clinically relevant acute GVHD (MAGIC II-IV) and acute GI-GVHD. These effects were attributed to the cytokine's immunomodulatory properties, particularly its promotion of regulatory T cells (T-reg) and maintenance of gut epithelial integrity [31,33,34,36]. TGF- β 2 FSMP may reduce GVHD-related morbidity by enhancing immune regulation and minimizing the loss of gastrointestinal barrier function, which is crucial in preventing bacterial translocation and subsequent systemic inflammation. This aligns with the broader literature on TGF- β 's protective role in inflammatory and autoimmune conditions [9–12,34–40]. By reducing GVHD incidence, TGF- β 2 FSMP contributes to improved overall clinical outcomes in patients who have undergone allo-HSCT.

4.3. Potential impact on microbiota composition

Modulen-IBD seems to have a certain effect also on microbiota composition: microbiota analysis in the post-transplant period, despite the low number of analyzed cases, highlights some differences in principal gut bacterial populations between the two groups. An appropriate supplementation of TGF- β 2 FSMP (Group A, TR $\geq$ 50 %) leads in gut microbiota to a higher relative abundance of beneficial families while an insufficient or sub-optimal TGF- β 2 FSMP supplementation (Group B, TR < 50 %) leads to greater relative abundance of proinflammatory families. In addition, the high relative abundances of *Lachnospiraceae* and *Clostridiaceae* found in Group A were negatively correlated with acute GI-GVHD occurrence, whereas the relative abundance of *Prevotellaceae* and *Bacteroidaceae* found in Group B was positively correlated with aGVHD [41,42]. In agreement with previously published papers, patients who developed acute GI-GVHD showed a depletion of commensal bacteria from the *Clostridiaceae* family that regulates butyrate production and contributes to improved survival and low incidence of aGVHD [8,43]. Moreover, Group A

(TR $\geq$ 50 %) showed a significant reduction in Firmicutes/Bacteroidetes (F/B) ratio after adequate intake of TGF- β 2 FSMP. The F/B ratio is widely recognized to affect intestinal homeostasis. Increased or decreased F/B ratio is considered as dysbiosis and can be related to several inflammatory bowel diseases, but in our study this finding needs to be further investigated due to the small number of patients [25]. These preliminary results show a trend that agrees with literature regarding pro- or anti-inflammatory gut populations, and that needs to be further explored and confirmed on a larger population [41,44].

5. Limitations and conclusions

Despite the promising findings, this study has some limitations. First, the non-randomized design and the division of patients based on their voluntary intake of TGF- β 2 FSMP may introduce selection bias and limit the generalizability of the results. Second, while the sample size is relatively large, the subgroup undergoing microbiota analysis was small ($n = 32$), which reduces the statistical power for detecting significant microbiota-related differences. Moreover, dietary habits outside the hospital setting and antibiotic exposures, which could have affected microbiota composition before transplantation, were not fully controlled. The follow-up period, although sufficient to assess short-term outcomes such as malnutrition and acute GVHD, may be too limited to evaluate long-term effects of TGF- β 2 FSMP on chronic GVHD and late infectious complications. Additionally, subjective factors, such as taste intolerance, impacted adherence to nutritional support and might have influenced clinical outcomes. Finally, mechanistic insights into how TGF- β 2 modulates immune responses and gut microbiota remain hypothetical and warrant further experimental validation. This study highlights the importance of nutritional support in patients undergoing stem cell transplantation on a prospective cohort of 192 real life consecutive transplanted patients. The use of TGF- β 2 FSMP in this setting should be investigated in an approved multicentric randomized controlled trial with the goal to optimize not only the nutritional aspects of allo-HSCT but also to investigate the microbiota-immune system axis and its influence on GVHD.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2025.06.044>.

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